Assignment 4: Comparative Analysis of Object Detection Frameworks: Single-Stage, Two-Stage, and Transformer-Based Models

(80 marks + 10 marks bonus)

COL7680/JRL7680/COL780: Introduction to Computer Vision

Due: 28 April 2026, 11:59 PM

1 General Instructions

Academic Integrity. This assignment must be completed individually. Plagiarism or academic dishonesty will be addressed per institute policy and may result in severe penalties.
Evaluation. All submissions will be followed by a demo session in which students must explain their approach and answer questions about their implementation.
Discussion. Conceptual discussion and troubleshooting are encouraged via the course Piazza forum: `https://piazza.com/iitd.ac.in/spring2026/col7680`. Do not share code or solutions.
Compute Resources. Students are free to use any external compute platforms such as Google Colab, Kaggle, or personal machines for training and evaluation. Please note that institute HPC resources will **not** be provided for this assignment. Ensure that your code is portable and can be executed with clear instructions.
Submission Report. Include a report covering: implementation approach and design decisions, challenges encountered and how they were resolved, deliverables and analysis asked to report, resources and references consulted, and any assumptions made.

2 Introduction

In this assignment, you will explore different object detection frameworks and systematically evaluate their performance under different modeling paradigms, training strategies, and data settings. You will begin with a single-stage detector (YOLO) to understand real-time detection and the trade-off between speed and accuracy. Next, you will work with a two-stage convolutional detector (Faster R-CNN with Feature Pyramid Network) to study the impact of different loss functions and the importance of multi-scale feature representations. Finally, you will experiment with a transformer-based detection model (Deformable DETR) and investigate how different fine-tuning strategies affect performance.

Building on these foundations, you will further examine the challenges of domain shift by adapting your best-performing model from synthetic training data to real-world data.

This will provide insight into the robustness and generalization capabilities of different detection architectures.

Through these tasks, you will gain hands-on experience in training, evaluating, and improving object detection models across varying object scales, categories, and domains. You will also develop an understanding of key design choices in modern detectors, including architectural differences (single-stage vs two-stage vs transformer-based), optimization strategies, and the role of components such as feature pyramids and attention mechanisms.

This assignment is designed to provide practical insights into how different detection models behave under diverse conditions, including scale variation, class imbalance, and domain shift. You will also analyze the effectiveness of fine-tuning in transformer-based detectors and explore solutions for adapting models to real-world scenarios.

Overall, the assignment will help you develop a comprehensive understanding of both classical and modern object detection techniques, along with their strengths, limitations, and real-world applications.

3 Dataset

The Benchmarking IR Dataset for Surveillance with Aerial Intelligence (BIRDSAI, pronounced "bird's-eye") is a long-wave thermal infrared dataset containing nighttime images of animals and humans in Southern Africa. The dataset allows for benchmarking of algorithms for automatic detection and tracking of humans and animals with both real and synthetic videos.

- **Download:** The data can be downloaded from the Labeled Information Library of Alexandria.
- **Annotation Format:** Dataset follow the MOT annotation format, which is a CSV with the following columns:

  <frame_number>, <object_id>, <x>, <y>, <w>, <h>, <class>, <species>, <occlusion>, <noise>

  - **class**: 0 if animals, 1 if humans
  - **species**: between -1 and 8 representing species below; 3 and 4 occur only in real data; 5, 6, 7, 8 occur only in synthetic data (note: most very small objects have unknown species; more details in paper [1])

4 Evaluation Protocol

Evaluate detection performance using mean Average Precision (mAP) with adaptations specific to the BIRDSAI Dataset. Since the dataset follows a Multi-Object Tracking (MOT) annotation format, each frame should be treated as an independent image for evaluation using Faster R-CNN, and mAP should be computed at a standard IoU threshold (e.g., 0.5 or COCO-style range, if implemented).

Unlike conventional COCO-style evaluation, object scale must be determined at the video level rather than per instance. For each video, compute the average bounding box area across all annotated objects and use this statistic to assign the video to one of three categories: small (average area < 200 pixels), medium (200–2000 pixels), or large (> 2000 pixels). All frames and objects within a video inherit its assigned scale category. Detection performance should then be reported as mAP separately for small-, medium-, and large-scale videos.

In addition to scale-based evaluation, compute class-wise mAP for the two target categories: Animals and Humans, using labels derived from the dataset annotations. Use IoU threshold (@0.5) and ensure that all assumptions regarding label mapping and evaluation details are clearly documented.

5 Task 1: Single-Stage Detection using YOLO[2] (15 Marks)

The objective of this task is to develop an understanding of fast, real-time object detection and to analyze the trade-off between detection speed and accuracy using a single-stage detection framework.

5.1 Instructions

BIRDSAI dataset will be used, with the analysis restricted to two semantic categories: **Animals and Humans**. The experimental setting follows a **Real** $\rightarrow$ **Real** paradigm, where both training and evaluation are conducted on real-world data.

A YOLO-based detector (eg, any recent YOLO version) should be employed for this task. The model must be trained on the specified dataset and subsequently evaluated to assess its detection performance. As part of the experiments, a baseline YOLO model should be trained first. Following this, a detailed performance analysis must be carried out by examining how the detector performs across different object scales—namely small, medium, and large objects—as well as across the two defined categories, Animals and Humans. This evaluation will help in understanding the strengths and limitations of single-stage detectors in terms of both efficiency and accuracy.

5.2 Deliverables

- **Code:** Provide a runnable script named **task1.sh** that executes the complete pipeline for this task. The script should follow the format:
  Example usage: `task1.sh <mode> <input_dir>`
  where **mode** can be train or evaluate. Ensure that the script is well-structured, reproducible, and includes all necessary steps such as data loading, model training, and evaluation.

- **Report:**

  - **Training Curves:** Plot and analyze the training behavior of the model. Include graphs showing **Loss vs Epoch** to illustrate convergence. Optionally comment on overfitting, underfitting, or stability of training.
  - **Quantitative Evaluation:** Present a detailed evaluation of detection performance using mean Average Precision (mAP) at IoU threshold 0.5. Organise results as shown in Table 1.

Table 1: mAP Comparison across Scales and Classes

Scale / Category	YOLO
SA (Small Animals)	
MA (Medium Animals)	
LA (Large Animals)	
Animals (Overall)	
SH (Small Humans)	
MH (Medium Humans)	
LH (Large Humans)	
Humans (Overall)	
Overall	

- **Qualitative analysis:** Visualize model predictions on sample images from the dataset. Include examples of correct detections as well as failure cases (e.g., missed detections, false positives). Briefly explain observed patterns in predictions.
- **Discuss** how the model performs across different object scales, particularly comparing large vs small objects.

6 Task 2: Two-Stage Detection using Faster R-CNN[3] (25 Marks)

The goal of this task is to perform a comprehensive evaluation of object detection performance using the Faster R-CNN framework, while developing an understanding of two key aspects: (1) the impact of different loss functions—Cross Entropy (CE) and Focal Loss (FL)—on detection performance, and (2) the importance of Feature Pyramid Networks (FPN) in handling objects at multiple scales. Students are expected to analyze how CE and Focal Loss influence class-wise and scale-wise performance, and how FPN contributes to improved detection, particularly for small and medium-sized objects.

6.1 Instructions

Use the BIRDSAI dataset and restrict the analysis to two semantic categories: **Animals** and **Humans**. The experiments should follow the **Real** → **Real** setting, where the model is trained and evaluated on real images.

Implement an object detection model using Faster R-CNN with a standard backbone such as ResNet-50, integrated with a Feature Pyramid Network (FPN). The inclusion of FPN is essential to enable multi-scale feature representation, allowing the detector to better handle objects of varying sizes.

Train two variants of the model to compare different classification strategies. The first variant should use Cross Entropy (CE), which treats all classes equally during training. The second variant should use Focal Loss (FL), which down-weights easy examples and focuses training on hard, misclassified examples, making it particularly effective in handling class imbalance and difficult samples.

6.2 Deliverables

- **Code:** Submit your complete code for both CE and Focal Loss training and evaluation, along with a bash script to run your code.
  Example usage: `task2.sh <experiment> <mode> <input_dir>`
- **Report:**
  - **Training Curves:** Display training curves (e.g., loss vs. epoch) for each experiment (CE and Focal Loss), and briefly comment on training behavior.
  - **Quantitative Evaluation:** Along with scale-based evaluation, calculate the class-wise mean Average Precision (mAP) for the two target categories: Animals and Humans. Use IoU threshold(@0.5) and clearly document assumptions regarding label mapping and evaluation protocol. Use the following table format:

    Table 2: mAP Comparison across Scales and Classes

Scale / Category	**FR-CE**	**FR-FL**
SA (Small Animals)		
MA (Medium Animals)		
LA (Large Animals)		
Animals (Overall)		
SH (Small Humans)		
MH (Medium Humans)		
LH (Large Humans)		
Humans (Overall)		
Overall		

    Perform qualitative analysis and provide a detailed discussion based on the best-performing Faster R-CNN model obtained from the above experiments (CE or Focal Loss).
  - **Qualitative Analysis:**
    - Provide visualizations of detection results, highlighting both successful detections and common failure cases for this task.
    - Visually compare the detection results of YOLO and Faster R-CNN + FPN on both easy and hard images.
      - **Easy images:** Images with clearly visible, large, and well-separated objects.
      - **Hard images:** Images containing small objects, occlusions, cluttered backgrounds, or multiple overlapping instances.
  - **Discussion:**
    - Compare the effects of CE and Focal Loss on detection performance, particularly in handling class imbalance and hard examples.
    - Analyze the contribution of the Feature Pyramid Network (FPN) in improving multi-scale detection.
    - Compare the speed vs accuracy trade-off between Faster R-CNN and YOLO (from Task 1), discussing differences between two-stage and single-stage detectors in terms of performance and efficiency.

7 Task 3: Evaluation and Fine-Tuning using Deformable DETR[4] (40 Marks)

The goal of this task is to evaluate and improve object detection performance using the Deformable DETR framework on the BIRDSAI dataset, while developing an understanding of how transformer-based detectors differ from and improve upon traditional CNN-based approaches such as Faster R-CNN and YOLO. In particular, students will analyze how DETR-based models eliminate the need for hand-designed components (e.g., anchor boxes, NMS) and leverage global attention for object detection. Additionally, the task focuses on understanding how different fine-tuning strategies impact model performance.

7.1 Instructions

Use the BIRDSAI dataset and restrict your analysis to the following two semantic categories: **Animals** and **Humans**. All experiments should follow the **Real** → **Real** setting, where both training and evaluation are performed on real images.

7.1.1 Evaluation of Pretrained Model

- Load a pretrained Deformable DETR model.
- Run inference on the dataset and perform qualitative analysis using visualization of detection results.
- Perform quantitative evaluation by generating precision-recall statistics at different thresholds and computing the mean Average Precision (mAP) for the pretrained model.

7.1.2 Fine-Tuning Strategies

Fine-tune the Deformable DETR model on the BIRDSAI dataset using appropriate training settings:

- Use the AdamW optimizer.
- Use the SetCriterion loss (as defined in DETR-based frameworks).
- Clearly document all hyperparameters (learning rate, batch size, number of epochs, etc.).

Perform the following experiments:

- **Experiment 1:** Fine-tune the full model (update all layers).
- **Experiment 2:** Fine-tune only the decoder.
- **Experiment 3:** Fine-tune only the encoder.

Save model checkpoints and compare detection performance across all experiments.

7.2 Deliverables

- **Code:** Submit your complete code for pretrained evaluation and all fine-tuning experiments, along with a bash script to run your code.
  Example usage: `task3.sh <experiment> <mode> <input_dir>`

- **Report:**

  - **Training Curves:** Display training curves (e.g., loss vs. epoch) for each fine-tuning experiment (Exp1, Exp2, Exp3), and briefly analyze convergence behavior.
  - **Quantitative Evaluation:** Along with scale-based evaluation, calculate the class-wise mean Average Precision (mAP) for the two target categories: Animals and Humans. Use IoU threshold (@0.5) and clearly document assumptions regarding label mapping and evaluation protocol. Use the following table format:

    Table 3: mAP Comparison for Pretrained and Fine-Tuned Models

Scale / Category	**Pretrained**	**Exp1**	**Exp2**	**Exp3**
SA (Small Animals)				
MA (Medium Animals)				
LA (Large Animals)				
Animals (Overall)				
SH (Small Humans)				
MH (Medium Humans)				
LH (Large Humans)				
Humans (Overall)				
Overall				

    Perform qualitative analysis and provide a detailed discussion based on the best-performing Deformable DETR obtained from the above experiments (Exp 1,2 and 3).

  - **Qualitative Analysis:**
    - * Provide visualizations of detection results, including both successful detections and failure cases, and comment on observed patterns.
    - * Visually compare the detection results of YOLO, Faster R-CNN + FPN, and Deformable-DETR models on both easy and hard images.
  - **Discussion:**
    - * Compare the effectiveness of different fine-tuning strategies (Exp1, Exp2, Exp3).
    - * Analyze how Deformable DETR performs across object scales compared to CNN-based detectors.
    - * Discuss why transformer-based detectors (DETR/Deformable DETR) can outperform traditional methods.
    - * Compare the speed vs accuracy trade-off of Deformable DETR with both Faster R-CNN (Task 2) and YOLO (Task 1).

8 Task 4: Domain Adaptation for Object Detection (Syn $\rightarrow$ Real) (Bonus 10 Marks)

The goal of this task is to study the impact of domain shift on object detection and evaluate how domain adaptation techniques can improve performance when training on synthetic data and testing on real-world data. Students will build upon their best-performing model from Task 1 (YOLO), Task 2 (Faster R-CNN + FPN with CE/WCE), or Task 3 (Deformable DETR), and adapt it to the Synthetic $\rightarrow$ Real (Syn $\rightarrow$ Real) setting. There will be **no partial marking** for this task.

8.1 Instructions

Use the BIRDSAI dataset and restrict your analysis to the following two semantic categories: **Animals** and **Humans**. In this task, the experimental setup changes to: **Training: Synthetic data** and **Evaluation: Real data**.

1. **Baseline Evaluation under Domain Shift:**
   Select the best-performing model from:

   - Task 1: YOLO (best configuration)
   - Task 2: Faster R-CNN + FPN (CE or WCE)
   - Task 3: Best fine-tuned Deformable DETR variant (Exp1, Exp2, or Exp3)

   Train (or reuse trained weights) using synthetic data only and evaluate directly on real data without any domain adaptation. This serves as the baseline to quantify the domain gap.

2. **Domain Adaptation Method:**
   Extend your chosen model by incorporating a domain adaptation strategy to address the domain shift between synthetic and real data. Instead of prescribing a fixed approach, you are encouraged to draw inspiration from existing research on domain-adaptive object detection. Some relevant references include:

   - Multiscale Domain Adaptive Yolo For Cross-Domain Object Detection[5]
   - Domain Adaptive Faster R-CNN for Object Detection in the Wild[6]
   - Dynamic Retraining-Updating Mean Teacher for Source-Free Object Detection (Deformable DETR)[7]

   You may either:

   - Reproduce a method from the above papers, or
   - Design and implement your own domain adaptation strategy by modifying your selected model.

   Your approach may involve changes at different stages of the detection pipeline, such as:

   - Feature extraction
   - Encoder/decoder representations (for DETR-based models)

- Loss functions (e.g., domain alignment losses)
- Training procedure (e.g., multi-domain or adversarial training)

3. **Important:**
Clearly describe how your method addresses the domain gap. Explain how it is integrated into your detection framework and justify your design choices with intuition and/or references to prior work.

8.2 Deliverables

- **Code:** Submit your complete code for baseline (Syn → Real) evaluation and domain adaptation experiments, along with a bash script to run your code.
Example usage:

```
task4.sh <model_type> <adaptation_method> <mode> <input_dir>
```

- **Report:**
  - **Training Details:** Provide all relevant training details, including hyperparameters, additional losses, architectural changes, or training strategies introduced for domain adaptation.
  - **Quantitative Evaluation:**
  Report scale-wise and class-wise mAP comparison:

Table 4: Domain Adaptation Performance (Syn → Real)

Scale / Category	Baseline	Adapted Model
SA (Small Animals)		
MA (Medium Animals)		
LA (Large Animals)		
Animals (Overall)		
SH (Small Humans)		
MH (Medium Humans)		
LH (Large Humans)		
Humans (Overall)		
Overall		

  Use IoU threshold(@0.5) and clearly mention any assumptions.
  - **Qualitative Analysis:** Provide a visual comparison between the baseline and adapted model. Highlight improvements in detection under domain shift as well as remaining failure cases.
  - **Discussion:** Provide insights on:
    * The effect of domain shift (Synthetic → Real) on detection performance
    * Effectiveness of the chosen adaptation method
    * Which object scales benefit the most (small vs medium vs large)
    * Challenges faced (e.g., training instability, domain gap severity) and how they were addressed

9 Grading Summary

Task	Marks
Task 1: Single-Stage Detection using YOLO	15
Task 2: Two-Stage Detection using Faster R-CNN + FPN	25
Task 3: Evaluation and Fine-Tuning using Deformable DETR	40
Task 4: Domain Adaptation for Object Detection(Syn $\rightarrow$ Real)	10 (bonus)
Total	**80 + 10 bonus**

Demo and Visualization Requirement. During the demo session, pipelines are **not required to train in real time**. Students are expected to present results using pre-trained checkpoints generated beforehand. Each task must include a minimal live visualization of detection outputs, such as predicted bounding boxes, class labels, and confidence scores overlaid on images.

Students must clearly demonstrate and compare results across different models, including; CE vs WCE (Task 1), Pretrained vs Fine-Tuned variants (Task 2) and Baseline vs Domain Adapted model (Task 3).

The visualization should highlight observable improvements or differences in detection performance, such as: improved localization (tighter bounding boxes), reduction in false positives/negatives, better detection of small or difficult objects, robustness under domain shift (Synthetic $\rightarrow$ Real).

Important: Students must provide a **Google Drive (or equivalent) link** containing all trained model checkpoints. The shared link must have **view-only access** with download permissions enabled and should **not be modified after the submission deadline**. Inference can be shown on a small subset of images and does **not need to be strictly real time**. Side-by-side comparisons or sequential visualizations are strongly encouraged Failure cases must also be shown and discussed. The demo should make it easy to qualitatively assess how each method improves upon the previous one.

References

[1] E. Bondi, R. Jain, P. Aggrawal, S. Anand, R. Hannaford, A. Kapoor, J. Piavis, S. Shah, L. Joppa, B. Dilkina, and M. Tambe, "Birdsai: A dataset for detection and tracking in aerial thermal infrared videos," in *WACV*, 2020.

[2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.

[3] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.

[4] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," 2021.

[5] M. Hnewa and H. Radha, "Multiscale domain adaptive yolo for cross-domain object detection," in *2021 IEEE International Conference on Image Processing (ICIP)*, p. 3323–3327, IEEE, Sept. 2021.

[6] Y. Chen, W. Li, C. Sakaridis, D. Dai, and L. Van Gool, "Domain adaptive faster r-cnn for object detection in the wild," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.

[7] T. L. B. Khanh, H.-H. Nguyen, L. H. Pham, D. N.-N. Tran, and J. W. Jeon, "Dynamic retraining-updating mean teacher for source-free object detection," 2024.